

Financial World-Model Runtime: Compiling Asynchronous Market Evidence for LLM Agents

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Abstract

Large language models and autonomous agents increasingly consume market evidence, yet raw financial feeds are asynchronous, intermittently missing, and quality-heterogeneous. Generative world models assume a usable observation stream; financial agents often lack that stream. We propose a *financial world-model runtime*: a point-in-time compiler that maps raw multi-band evidence into an AI-consumable world state with explicit quality scores and abstention. The primitive is an epistemic observation $O_{j,t}=(x,\tau,q,g,r)$; a compilation operator Π_t yields $\blacksquare_t^{\text{AI}}$; WMI/ACWMI gate actions. Empirically, on a cryptocurrency PIT panel, vintaged macro and stablecoin-flow *content* changes a transparent consumer rule relative to same-input momentum ($\Delta\text{CE } 0.334, p=0.034$), with LOBO losses dominated by content. A Compiled-versus-Raw protocol tests the agent interface. **Thesis**: trading metrics validate that the compiled world has content—they do not proclaim a strategy Holy Grail.

Keywords: world models; LLM agents; information-set compilation; selective prediction; point-in-time; cryptocurrency

1. Introduction

AI agents that analyze or trade markets do not observe a clean panel. Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks. Literatures on ML asset pricing and on generative world models (Ha & Schmidhuber; Dreamer; JEPA-style proposals) largely assume observations are already aligned. For LLM and tool-using agents, the prior systems problem is different: *compile* asynchronous evidence into a world state that can be read, refused when thin, and acted on when ready.

Thesis. We build a *financial world-model runtime*—an observation-side world model that compiles raw market evidence into a quality-tagged state for AI consumers. Trading statistics show nonempty content; they are not unconditional strategy alpha.

Contributions. (1) *Semantics*: epistemic observations, Π_t , reconstruction bound, WMI/ACWMI, world-conditional abstention. (2) *Runtime*: PIT multi-band system, quality-tagged bundles, availability shocks, Compiled-versus-Raw protocol. (3) *Validation*: PIT content yields significant OOS CE vs same-input momentum; LOBO is content-dominated.

2. Related Work

World models in AI. World models learn compact states and often dynamics for imagination/control. We address the complementary *observation-side* problem in markets: PIT-safe state construction with explicit quality before any dynamics model or LLM policy.

LLM agents / selective prediction / PIT. Tool-using LLMs need a compiled world interface. Selective classification justifies abstention under noise; we tie abstention to world quality. Macro vintages and crypto fragmentation motivate multi-band compilation.

3. Financial World-Model Runtime

Definition (epistemic observation). $O_{j,t}=(x,\tau,q,g,r)$: value, latest-available time, quality, main-view gate, semantic role.

Definition (compilation). $W_t=\Pi_t(\blacksquare_t^{\text{raw}})$ with $\Pi_t=B_tM_tA_t$; $\blacksquare_t^{\text{AI}}=\sigma(W_t)$. WMI $_t=B_tU_tH_t$ scores world quality; ACWMI provides regime-conditional gating.

$$\blacksquare S_t - S_t \leq C_1 \cdot \text{delay} + C_2 \cdot \text{noise} + C_3 \cdot \text{missingness}.$$

Prop. 1 (compilation ≠ feature expansion). Enlarging raw span without Π need not enlarge usable $\blacksquare^{\text{AI}}$. **Prop. 2 (world-conditional abstention)**. If all non-abstain actions exceed $c_{\text{abs}}(W)$, abstain. **Def. (LOBO)**. Band value = content channel + gating channel.

Remark (scope). We do *not* claim a learned $p(s_{t+1}|s_t,a_t)$. This paper is a *state compiler + abstention runtime*; generative dynamics are future work. Do not read this as Dreamer-for-markets.

4. System

Anonymized prototype: multi-band collectors → vintage stores → BandPIT under previous-close clock (payoff r_t uses info at $(t-1)$ 23:59) → readiness/WMI/ACWMI → availability shocks O_t → quality-tagged bundles → Compiled-versus-Raw consumers (frozen prompts, $T=0$, actions include abstain). Durable bands: {exchange, macro, alternative}.

Raw feeds → Π_t / quality → World bundle W_t → LLM or rule consumer → {trade, abstain}

Figure 1 (schematic). Observation-side financial world-model runtime.

5. Experiments

Setup. PIT ~399 days, 10 liquid crypto names, prev-close clock, IS/OOS 200/200. Pre-specified contrast: mechanism (band content) – momentum. Bootstrap $n=999$, block=5; stationary confirmation.

RQ1 Content. Mechanism Sharpe/CE 0.767/0.132 vs momentum 0.101/–0.202; $\Delta\text{CE}=0.334$ ($p=0.034$; stationary 0.022). Relative gap survives 10–25 bps costs; absolute CE is cost-fragile.

RQ2 LOBO. Dropping macro or alternative collapses to momentum ($\Delta\text{CE}=-0.334$); content share=1.0 under ungated mechanism (gating residual 0). Identifies content channel; nontrivial gating needs denser archives.

RQ3 Interface. Compiled-versus-Raw offline mocks keep the protocol executable; $\text{EAR}\approx 1$. No live LLM alpha claimed.

RQ4 Audit. 2017–2026 without vintaged band archives: no hidden return-rule edge over momentum; unsigned proxies can fire without recreating PIT CE. Gains load on compiled PIT content.

6. Limitations

Sparse bands beyond exchange/macro/alternative; support-matched WMI thresholds required; absolute CE cost-fragile; consumer tests are protocol-first (mocks); observation-side runtime $\neq$ generative dynamics WM; no menu-wide claim versus always-long.

7. Conclusion

Financial AI agents need a world to read. We formalize and implement a world-model runtime that compiles asynchronous evidence into an analyzable, abstention-aware, tradeable state, and show that compiled band content is economically nonempty under a disciplined PIT design. Generative dynamics and live multi-vendor LLM consumers are natural next layers on the same interface.

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